

Protocol



A workflow for statistical analysis and visualization of microbiome omics data using the R microeco package

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Abstract

The increasing complexity of experimental designs and the volume of data in the microbiome field, along with the diversification of omics data types, pose substantial challenges to statistical analysis and visualization. Here we present a step-by-step protocol based on the R microeco package (<https://github.com/ChiLiubio/microeco>) that details the statistical analysis and visualization of microbiome data. The omics data types shown consist of amplicon sequencing data, metagenomic sequencing data and nontargeted metabolomics data. The analysis of amplicon sequencing data specifically involves data preprocessing and normalization, core taxa, alpha diversity, beta diversity, differential abundance testing and machine learning. We consider various data analysis scenarios in each section to exhibit the comprehensiveness of the protocol. We emphasize that different normalized data produced by various methods are selected for subsequent analysis of each part based on the best analytical practices. Additionally, in the differential abundance test analysis, we adopt parametric community simulation to enable the performance evaluation of various testing approaches. For the analysis of metagenomic data, the focus is on how bioinformatic analysis data are read and preprocessed, which refers to the major usage differences from amplicon sequencing data. For metabolomics data, we mainly demonstrate the differential test, machine learning and association analysis with microbial abundances. To address some complex analyses, this protocol extensively combines different types of methods to build an analysis pipeline. This protocol is more comprehensive and scalable compared with alternative methods. The provided R codes can run in about 6 h on a laptop computer.

Key points

- This step-by-step protocol details the use of the R microeco package for the statistical analysis and visualization of microbiome data, covering omics data types such as amplicon sequencing data, metagenomic sequencing data and nontargeted metabolomics data. This protocol extensively combines different types of methods to build an analysis pipeline.
- This protocol is more comprehensive and scalable compared with alternative methods.

Key references

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Introduction

Tools for the statistical analysis and visualization of microbiome data

Microorganisms are ubiquitous and have pivotal roles in various habitats, such as the human gut¹, plant rhizosphere soil² and oceans³. Microbial communities typically comprise hundreds to thousands of species and exhibit substantial spatial and temporal variability. Over the past two decades, high-throughput sequencing technologies, especially amplicon sequencing, have driven microbiome research into a phase of explosive growth, elevating our understanding of microbial communities to unprecedented levels. Bioinformatics tools for amplicon sequencing data analysis have evolved over the years, resulting in mature software systems such as QIIME2 (ref. 4), which has become one of the most widely used. QIIME2 integrates many renowned methods, such as DADA2 (ref. 5) for generating amplicon sequence variants (ASVs) and machine learning algorithms for taxonomic classification. In downstream data analysis, the R language⁶ has emerged as the most commonly used platform for statistical analysis owing to its numerous advantages (for example, open-source nature, rich package resources). These downstream analyses involve a wide range of statistical and visualization methods, many of which are challenging to use and have specific application scenarios⁷. Mastering these methods requires interdisciplinary expertise in biology, ecology, statistics and programming, which imposes a heavy learning and operational burden on users. This challenge has led to the development of R packages designed to facilitate more convenient and comprehensive statistical analysis and visualization of microbiome data, as reviewed by Wen et al.⁸. Among these packages, the *microeco* package⁹, which we developed and maintain (<https://github.com/ChiLiubio/microeco>), has gained widespread adoption. The primary goal of the *microeco* package is to streamline the statistical analysis of microbiomes and seamlessly integrate it with visualization, while maintaining robust functionality. Compared with other R packages for microbiome data analysis (for example, *phyloseq*¹⁰), the *microeco* package offers several unique advantages, including a modular design based on the R6 class¹¹, a diverse array of functions and methods, and seamless integration of statistical analysis and visualization.

Development and use of *microeco*

Software maintenance is crucial for improving the quality of software^{12,13}. Since its initial release in 2020, the *microeco* package has been updated numerous times, introducing a multitude of new features. Over the past 12 months, the package has been downloaded over 35,000 times from the Comprehensive R Archive Network (CRAN). The main methods in this protocol originate from the maintenance process of the *microeco* package after its initial publication. The new features added to the *microeco* package primarily focus on the following aspects: (1) introducing new classes to address additional analysis needs (for example, the *trans_norm* class for data normalization and *trans_classifier* class for machine learning model analysis); (2) providing more method options within functions, allowing users to select different algorithms (such as additional differential test methods in the *trans_alpha* and *trans_diff* classes); (3) adding functions to classes to implement various types of analysis (for example, the *plot_scatterfit* function in the *trans_env* class) and (4) incorporating more parameters in functions to handle more complex data structures (for example, the *by_group* parameter in the new function of *trans_alpha* class for nested group comparisons). Some of these methods hold notable value but may not be widely recognized by researchers in certain fields. For instance, researchers who lean towards studying the ecology of microbial communities in environmental samples often prioritize diversity analysis, while paying less attention to differential abundance testing methods, particularly those designed for analyzing differences in ASVs or operational taxonomic units (OTUs) (ASVs/OTUs). Researchers unfamiliar with advanced statistical analyses may not be well-versed in methods related to generalized linear mixed-effects models or machine learning models. Similarly, those with limited experience in high-throughput sequencing data may find themselves unfamiliar with normalization methods applicable to amplicon sequencing data, leading to potential confusion when selecting appropriate approaches. Additionally, due to individual data analysis habits or the adoption of traditional

approaches within research groups, many analytical workflows may remain underutilized. For example, the use of centered log-ratio (CLR) normalization combined with Euclidean distance to derive the Aitchison distance matrix¹⁴ is not widely known. Some unique features of the *microeco* package also deserve broader attention, such as the integration of different classes to address specific problems, as partially described in previous research¹⁵. The combination of these methods requires a certain level of expertise, which can be difficult for beginners without a thorough understanding of the *microeco* package. However, with access to a substantial number of examples, these techniques can be quickly mastered. These considerations motivate the development of this protocol, which provides a set of best-practice strategies for addressing various research questions, combining amplicon sequencing data normalization with subsequent statistical analysis and visualization.

Integration of multiomics approaches in *microeco*

Owing to the immensely complex nature of microbial consortia, the integration of multiomics approaches (for example, amplicon sequencing, metagenomics, metatranscriptomics and metabolomics) holds crucial value for elucidating the mechanisms underlying relevant research questions. Different meta-omics techniques in the microbiome field are tailored to address distinct ecological or biological questions related to microbial composition, function and activity¹⁶. Among these methods, the analysis of metagenomic and metatranscriptomic data is particularly complex, especially metagenomic sequencing data, which provides comprehensive insights into various aspects (for example, structure and abundance) of meta-genomes and involves diverse bioinformatics workflows. Bioinformatic analysis of metagenomic sequences can be generally divided into two major categories: (1) analysis focused on contigs, binning and the functional characterization of draft genomes and (2) analysis focused on the abundance of genes, metabolic pathways or species. A suite of complementary tools for these analyses has matured and become increasingly powerful. For example, the HUMAnN3 software¹⁷ is widely used for analyzing metagenomic and metatranscriptomic sequencing data to quantify the abundance of genes and metabolic pathways. Compared with other tools for functional profiling of metagenomic data, its key advantage lies in its ability to provide functional insights at the species level. The Kraken2 software¹⁸ aims to classify metagenomic sequences to obtain the abundances of taxa across various taxonomic levels. Kraken2 examines the *k*-mers within a query sequence and uses the information within those *k*-mers to query a database. This database maps *k*-mers to the lowest common ancestor of all genomes known to contain a given *k*-mer. The abundances of genes, metabolic pathways or taxa in metagenomic analysis results share some common features with the ASV abundances obtained from amplicon sequencing, such as both being sample-feature abundance tables and having hierarchical annotation information. However, the files produced by bioinformatics tools for amplicon sequencing data are relatively simple, with minor difference in file formats but similar content. By contrast, the types of abundance files produced by bioinformatics platforms for metagenomic analysis vary greatly. To unify subsequent analysis processes, it is necessary to design reading functions tailored to the characteristics of the files produced by these software. Therefore, we developed the *file2meco* package¹⁹, which is specifically designed to read various types of data and generate microtable objects to enable seamless conversion of data from software such as HUMAnN3 and Kraken2 into formats compatible with the *microeco* package. We have also optimized several functions within the *microeco* package to accommodate these more complex data types. The integration of multiple omics methods increases the complexity and workload of data analysis, posing challenges for beginners in bioinformatics and R programming. To address this, we have included examples of metagenomic and metabolomic data in this protocol to demonstrate common and critical analyses, while also highlighting key differences compared with amplicon sequencing data analysis.

Applications

The multiomics data types used in this protocol are relatively common in microbiome-related research. The various methods demonstrated, which are of different types, also have high universality across different research fields. Therefore, this protocol has broad applicability.

For instance, researchers in environmental microbiology can apply it to analyze microbial communities in soil, water and other environments, offering insights into ecosystem dynamics and the effect of environmental changes. In agriculture, it aids in exploring soil–plant–microbiome relationships under various experimental treatments. It can also be used in human microbiome studies to investigate the community changes under specific conditions such as diseases. The protocol supports multiple analytical objectives: assessing community structure through alpha and beta diversity analysis, identifying differentially abundant taxa, building prediction models, exploring microbial functions and conducting correlation analyses with different omics data to reveal ecological relationships. The workflows are adaptable to longitudinal, case–control or intervention studies, ensuring broad utility in hypothesis-driven and exploratory research.

Example data for the demonstration of the protocol

We demonstrate the protocol using data obtained from a two-factor field crop experiment concerning cropping and fertilization treatments²⁰. In agriculture, cropping and fertilization practices can largely influence soil nutrient contents and microbial communities, thereby affecting the rhizosphere and endophytic microbial communities of crops, as well as crop health. Based on this background, we investigate how cropping and fertilization treatments affect the structure, function and metabolic characteristics of microbial communities in bulk soil and maize rhizosphere soil. We hypothesize that the mechanisms by which cropping treatments and fertilization treatments affect soil and maize rhizosphere microbiota are fundamentally different. Cropping treatments include soybean–maize rotation (RC) and continuous maize cropping (CC). Fertilization treatments consist of CK (no fertilizer), NPK (inorganic fertilizer application) and NPKS (NPK + straw amendment). Thus, there are a total of six treatment groups: RC–CK, RC–NPK, RC–NPKS, CC–CK, CC–NPK and CC–NPKS. Each group has five biological replicates. Amplicon sequencing data cover three compartments: bulk soil, rhizosphere soil and the root (for profiling endophytes). Metagenomic sequencing and untargeted metabolomics data include two compartments: bulk soil and rhizosphere soil.

Overview of the protocol

This protocol describes a set of modularized approaches for the statistical analysis and visualization of amplicon sequencing data, metagenomic sequencing data and metabolomic data (Fig. 1). In designing the entire procedure, we paid special attention to the combination of methods, as this is a common scenario in microbial community data analysis. The combination of methods highlights the flexibility and efficiency of the software package, thereby providing users with more options for their analysis processes. Each section of the workflow is organized according to the type of analysis content and the protocol is divided into the following stages to facilitate understanding and operation: ‘Stage 1: install R packages’, ‘Stage 2: generate microtable object for amplicon sequencing data’, ‘Stage 3: core microbiome analysis’, ‘Stage 4: alpha diversity analysis’, ‘Stage 5: beta diversity analysis’, ‘Stage 6: differential abundance test’, ‘Stage 7: machine learning models’, ‘Stage 8: metagenomic data analysis (HUMAnN3 software)’, ‘Stage 9: metagenomic data analysis (Kraken2/Bracken software)’ and ‘Stage 10: metabolomic data analysis’. Specifically, Stage 1 corresponds to the first step in Fig. 1, which is used to install the various R packages necessary for this protocol. Stages 2–7 pertain to the analysis of amplicon sequencing data, involving crucial steps for data reading and preprocessing as well as commonly used data analysis methods in microbial ecology. Stages 8 and 9 are designed to demonstrate the reading and subsequent analysis of result files generated by two commonly used metagenomic bioinformatics analysis tools. Finally, Stage 10 is dedicated to showcasing various analyses of metabolomic data.

We selected one representative case study with multiple treatment factors and environmental metadata to demonstrate the various applications of the methods. For amplicon sequencing data, owing to variations in sequencing depth across samples it is necessary to normalize the data to avoid the effect of sequencing depth on sample comparisons (Box 1). As a result, each analysis step is paired with an appropriate data normalization method based on empirical knowledge. For example, resampling methods are used in core ASV analysis and

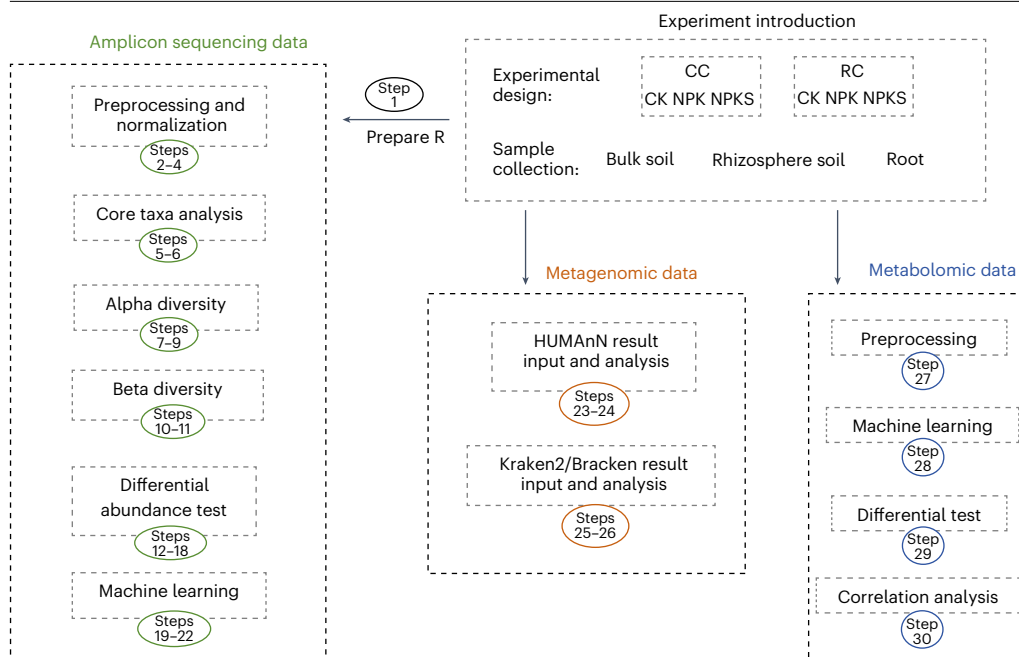


Fig. 1 | The framework of the protocol. After setting up the R language environment (Step 1), start with the preprocessing and normalization of bioinformatics analysis results of amplicon sequencing data (Steps 2–4). Then, proceed with core microbiome analysis (Steps 5–6), alpha diversity analysis (Steps 7–9), beta diversity analysis (Steps 10–11), differential abundance test analysis (Steps 12–18) and machine learning analysis (Steps 19–22). Shotgun metagenomic sequencing data analysis is conducted separately for HUMAnN3 (Steps 23–24) and Kraken2/Bracken (Steps 25–26). Nontargeted metabolomic data analysis includes data preprocessing (Step 27), machine learning analysis (Step 28), differential abundance test (Step 29) and correlation analysis with microbial taxa (Step 30).

alpha and beta diversity calculations, and scaling factor normalization methods are used in differential abundance testing. In this protocol, visualization in each section is typically accompanied by specific statistical analysis. However, the application of graph adjustments is beyond the scope of this protocol.

Advantages and limitations compared with alternatives

Compared with other protocols based on online tools²¹ or R packages⁸ in this field, this protocol offers several distinct advantages: (1) it provides a comprehensive overview of a wide range of methods based on the R microeco package, making it particularly beneficial for beginners to engage in systematic learning of statistical analysis and visualization of microbiome omics data. (2) It provides a variety of well-established algorithms of the same type but with certain differences that are used in some operations (for example, differential abundance test). This helps users to quickly become familiar with these methods, reducing the learning curve and improving usability. (3) Complex tasks in data analysis can be addressed by combining different classes within the microeco package. This enables users to better understand the design philosophy and advantages of the package (for example, high modularity and seamless integration of statistics and visualization) and apply it flexibly to their own data. (4) The protocol covers multiomics data (particularly metagenomic and metabolomic data), offering diverse analytical procedures to suit different research needs. (5) Parametric community simulation is included to expand the application scenarios of the methods. Therefore, this protocol not only enables newcomers to quickly learn and master the necessary skills for analyzing multiomics data in microbiome research but also provides a wealth of methodological resources for researchers with prior experience in microbiome data analysis. It is worth noting that although many methods described in this protocol may not be directly selected by practitioners for their own data analysis, they are of great value as part of a

BOX 1

Data normalization

In the statistical analysis of amplicon sequencing data, the first, and also most controversial, step is data normalization. Among the various considerations, sequencing depth is the primary factor to address. In sequencing results, different samples exhibit unequal sequencing depths (library sizes), which can greatly affect downstream analyses, such as diversity calculations and differential abundance tests. Additionally, the count data generated by sequencing reflects relative abundance rather than absolute abundance within the community. This can lead to a phenomenon where an increase in the absolute abundance of one taxon results in observed changes in the abundance of other taxa, a problem known as compositionality³⁸. Amplicon sequencing data also suffer from sparsity, with a high prevalence of zero values that may affect the applicability of certain statistical models. Common data normalization methods can be categorized into several types. The first is rarefaction, which resamples all samples to a uniform sequencing depth by randomly subsampling sequences from samples that meet a specified threshold⁷¹. The second category is scaling normalization methods, which use scaling factors to adjust the observed counts³⁸. Some of these methods were originally designed to convert relative abundance data into absolute abundance estimates^{38,54}, including those developed for transcriptome data such as DESeq2³⁴ and edgeR³⁵. Although TSS falls under the category of scaling normalization, it is a proportion-based correction method and does not address the issue of compositionality. Another category of normalization methods is specifically designed to tackle the compositionality problem, such as CLR and additive log-ratio transformations²⁹.

Despite considerable advancements in the development of normalization methods, researchers have yet to reach a consensus on the optimal approach, with ongoing debates regarding the suitability of various normalization techniques. For instance, the rarefaction method is often criticized for discarding a substantial portion of sequence data, which can reduce the statistical power of differential abundance tests. However, studies benchmarking beta diversity at the community level have found that rarefaction and proportion-based methods more accurately reflect true community beta diversity compared with other normalization approaches⁷². On the other hand, many differential abundance test methods have been shown to exhibit high FDRs under various simulated data conditions^{55,61,73}. The TSS method is also frequently criticized for failing to resolve the compositionality issue³⁸, as the relative abundances derived from this method often deviate significantly from the true abundances in the samples, leading to higher false-positive rates in differential abundance tests. However, an often overlooked but crucial point is that relative abundance, from the perspective of maximum likelihood in the multinomial distribution⁷⁴, represents the occurrence probability of taxa (at any taxonomic level) within a community. A higher relative abundance of a microbial group suggests a greater competitive advantage in maintaining its abundance during community dynamics, such as in resource competition among species. In conclusion, there is no universally perfect normalization method that suits all downstream analyses. The choice of normalization should be guided by the specific context and goals of the data analysis.

comprehensive learning framework. These methods help users to familiarize themselves with the methodological landscape of microbial ecology.

Of course, this protocol has some limitations. First, beginners may encounter difficulties in understanding the details of certain approaches, particularly those that have been developed over many years and require substantial background knowledge. However, these challenges are unlikely to hinder the execution of the processes demonstrated in the protocol. Second, this protocol assumes that users have some basic knowledge of the R language and an understanding of the architecture of the microeco package, particularly its reliance on the R6 class¹¹. This adds a learning burden compared to some web-based workflows²¹ that do not require prior R language experience.

Level of expertise required

A basic understanding of the R language is required to comprehend the general operations within this protocol. This includes tasks such as importing data into a data frame format, performing data frame filtering operations, adjusting plotting details using the theme function from the ggplot2 package, and saving tables and figures to local files. As there is a wealth of introductory material on the R language available online, individuals with insufficient R training can quickly familiarize themselves with it by practicing with these resources or systematically learning the basics through books. Some books have been mentioned in other protocols²², and others are also highly suitable for beginners^{23–25}. In addition to the basics of the R language, users also need to have a general understanding of the design of the microeco package⁹,

including the purpose and methods of each class. These concepts can be quickly grasped by referring to the published literature or the package's help documentation. Among these requirements, the most fundamental aspect is that users need to understand the creation of R6 class objects and their manipulation methods. These differ markedly from the S4 class objects used by some R packages, such as phyloseq. Information regarding the R6 and S4 classes can be obtained through online resources or the free book²⁶. In addition to proficiency in the R language, users should also have a solid understanding of microbiome experiments, including experimental design types²⁷, high-throughput sequencing²⁸, and bioinformatics and statistical analysis methods²⁷. We have provided explanations of some key terms related to this protocol to help beginners to quickly become familiar with the concepts of ecological and statistical methods (Box 2).

Experimental design

Factor settings of experimental control group

The example data of this protocol come from a two-factor treatment experiment and an additional sampling factor from different root compartments. Users with either single-factor or multifactor experimental designs can directly apply the provided steps. It is important to note that the setting of group factors in the data preprocessing section is crucial. The first element should be set as the control group, as demonstrated in the procedure (Step 3).

BOX 2

Glossary of statistical and biological descriptors

Glossary	Description
ASV	The amplicon sequence variant (ASV) is generated on the basis of some bioinformatic methods from amplicon sequencing data. ASV methods distinguish sequence variants differing by as little as one nucleotide and do not cluster sequences like the operational taxonomic unit (OTU) method, such as at a 97% sequence similarity threshold
Core microbiome	The species/ASV/OTU with high occurrence frequency and abundance across samples
Differential abundance test	A test of the significant differences in the abundances of taxa across groups. In microbiomes, commonly used statistical methods differ largely in terms of the data normalization and underlying hypothetical statistical distributions
False discovery rate (FDR)	The proportion of samples declared as significant by a given test that in fact belong to true null hypothesis (no effect). FDR is defined as: $FP/(TP + FP)$. Please refer to the 'FN/FP/TN/TP' section for the definitions of FP and TP
Feature selection	The process that reduce the number of input features in a dataset by selecting the most informative subset of features. Feature selection enhances model performance by eliminating redundant features, thus preventing overfitting and making the model more interpretable
FN/FP/TN/TP	FN (false negative): a variable that does not conform to the null hypothesis of no effect but is mistakenly declared to conform by a statistical test. FP (false positive): a variable, which actually belongs to the null hypothesis of no effect, is considered to be significant by a statistical test. TN (true negative): a variable correctly tested as negative. TP: (true positive): a variable correctly tested as positive
Generalized linear mixed models	Statistical models that assume errors from the exponential family and also include both fixed and random effects; predicted values are determined by discrete and continuous predictor variables and by the link function (for example, logistic regression, Poisson regression, beta regression)
Linear mixed models	Statistical models that assume normally distributed errors and also include both fixed and random effects
Normalization	A data preprocessing step that eliminates the effects of sequencing depth to make profiles of different samples comparable. The methods commonly used in microbiome field includes rarefaction and the scaling factor methods (such as total sum scaling and centered-log ratio normalization)
Ordination	Graphical representation of the similarity of samples and/or species units in a low-dimensional space. Commonly used ordination methods in microbial ecology includes exploratory methods (unconstrained ordination, for example, PCA and PCoA) and hypothesis-driven methods (constrained ordination, for example, RDA)
Parametric community simulation	Using statistical model to parameterize real-world microbial community profiles and to simulate new profiles with known structure for methods evaluation
ROC curve/PR curve	A type of curve demonstrating the model performance at different thresholds of classification probability in machine learning. ROC curve: receiver operating characteristic curve, which is used to show the relationship between true-positive rate (also called sensitivity or recall and defined as $TP/(TP + FN)$) and the false-positive rate (defined as $FP/(TN + FP)$; equal to $(1 - \text{specificity})$). PR curve: precision-recall curve, used to show the relationship between precision (defined as $TP/(TP + FP)$) and recall

This ensures accurate calculations in subsequent analyses (such as linear regression methods) and reasonable presentation of results. If some methods in the workflow do not analyze based on specific experimental factors, it indicates that these methods have strong versatility and are not dependent on the experimental design.

Replicates

For experiments with different treatment groups, we recommend collecting at least four replicate samples for each group. The reason is that a sufficient number of replicates is necessary for the numerous statistical analysis methods in the protocol, such as pairwise comparisons between groups in alpha diversity and differential abundance analysis. For some sampling types, such as environmental samples, there may not be strict biological replicates, but it is still advisable to ensure that each group has at least four samples. For core microbiota analysis and correlation analysis, it is suggested that each group (or all samples if no grouping is applied) should include at least ten samples to ensure the reliability of the results. The ordination analysis in beta diversity and the visualization part of metagenomic data do not have specific requirements for sample replication.

Choice of data normalization method

For amplicon sequencing data, data normalization is a necessary step before using certain data analysis methods. Some differential abundance test methods incorporate data normalization techniques, but many other common analyses require addressing the effect of sequencing depth first. In some data analysis cases, issues of data compositionality and sparsity also need to be considered. Although rarefaction is often criticized because it results in data loss and lowers statistical power, it remains the most frequently used method for normalizing data in diversity-related analyses. This protocol also demonstrates other different normalization methods, including CLR^{14,29}, robust CLR³⁰, cumulative sum scaling (CSS)³¹, geometric mean of pairwise ratios (GMPR)³², Wrench³³, median ratio of counts relative to geometric mean (DESeq2)³⁴, trimmed mean of *M* values (TMM)³⁵, relative log expression and total sum scaling (TSS, also known as Proportion). Rarefaction is useful in the assessment of alpha diversity. Rarefaction, TSS and CLR/robust CLR approaches are suitable for beta diversity analysis, as they preserve the relative abundance relationships among features. The TSS method is also commonly used to analyze relative abundance at higher taxonomic levels (without considering compositional effects), such as in the visualization and differential analysis of relative abundance (for example, LEfSe method³⁶ and beta regression³⁷) and in machine learning applications. Other normalization methods are primarily used for differential abundance analysis of ASVs/OTUs³⁸, and they tend to mitigate the compositional effects, as well as the issue of sequencing depth. Methods such as DESeq2, TMM and relative log expression methods, originally developed for transcriptomic data, assume that the absolute abundance of most features remains constant. These methods are suitable for ASV/OTU abundance data under certain conditions: (1) experimental treatments minimally affect the absolute abundance of most taxa and (2) the dataset contains few zero values. For sparse ASV/OTU data, Wrench or CSS may be more appropriate than DESeq2 or TMM. Additionally, in cases of high count variability in datasets, alternative methods such as GMPR should be considered instead of the aforementioned approaches. Considering that the statistical assumptions underlying these methods for differential abundance analysis of ASVs/OTUs may not be satisfied in the users' amplicon sequencing data⁷, comparisons among different methods should be conducted to select the best method when necessary (as demonstrated in Steps 16–18).

Choice of beta diversity analysis method

In the beta diversity analysis, some indices for calculating community distances (for example, Bray–Curtis dissimilarity³⁹) can be affected by the sequencing depth of the samples. To reduce the effect of sequencing depth, normalized data can be used to calculate these indices. In the `cal_betadiv` function of `microtable` object, the Bray–Curtis dissimilarity is calculated based on the abundance table, while the Jaccard distance is calculated using binary presence/absence data by default. Users can modify the parameters according to their needs. When it is necessary to

consider the phylogenetic relationships among species in some experiments, UniFrac distance indices can be used⁴⁰. The Jaccard and unweighted UniFrac distance are more applicable in cases in which there is a large variation among samples and the abundance information of taxa is not important. Aitchison dissimilarity can be calculated using the original data as there is data normalization (CLR method) inside the calculation process. When the sample size is large or the data variation is high (such as data under certain gradients), the effect of ordination analysis may not be satisfactory (for example, when some points deviate too far from the others). In such cases, it is possible to try using non-metric multidimensional scaling instead of principal coordinates analysis (PCoA), or detrended correspondence analysis (DCA) instead of principal component analysis (PCA). For PCA or DCA, square root transformation can also be considered.

Choice of differential abundance test method

The differential abundance test aims to identify taxa that exhibit significant changes in abundance across different groups, thereby assisting in explaining the community assembly patterns between groups. In terms of experimental setup, the most common scenario is two groups under a single-factor setting, namely the treatment group and the control group. However, there are many experiments with three or more groups under one factor. Some studies may have multiple factors or need to consider random effects. For different types of experimental designs, users can modify the corresponding steps and parameters to fit their own data. Some of the demonstrated methods also support the input of continuous variables. In terms of taxonomic information, some studies focus on the most refined taxonomic level of species or ASV/OTU, and some may tend to perform a differential test on higher taxonomic levels (for example, genus and phylum). The choice of calculation methods for differential abundance test for ASV/OTU has been discussed in the 'Experimental design' section entitled 'Choice of data normalization method'. Some traits of microorganisms are conserved to a certain degree in phylogenetic relationships, and different traits can vary markedly, resulting in distinct niches of microorganisms. Such differences in the niches can often be reflected in the abundances of higher taxonomic groups under varying external conditions. Therefore, conducting differential abundance test at higher taxonomic levels helps to analyze the connections between experimental factors and microbial taxonomy (or traits). The choice of taxonomic level in the analysis depends on the biological question the experiment aims to address.

Choice of machine learning method

Machine learning has been widely applied in microbiological research, such as predicting diseases based on gut microbiota⁴¹ and predicting soil health based on soil microbiota⁴². A key reason for its popularity in microbial ecology is that machine learning approaches have minimal requirements for experiment design and data structure and are suitable for processing relatively complex data, such as microbiomes (where features far outnumber samples). Although some studies have provided recommendations for applying machine learning methods in the microbiome field⁴³, the actual operation remains extremely challenging for many users. This protocol provides computational steps suitable for both categorical and continuous variables, thereby accommodating a wide range of experimental design types. The choice of a specific taxonomic level should depend on the research question, as discussed in the previous section and a previous article⁴³. If the user is unsure which machine learning algorithm to choose, comparing several commonly used algorithms—as demonstrated in Step 21—is a good approach.

Materials

Equipment

- A computer
 - ▲ **CRITICAL** The current process does not have special requirements for computer performance; ordinary desktop or laptop computers are sufficient to run it. However, the available memory should be at least 4 GB.

Programming environments

- R language software (<https://cran.r-project.org/>)
▲ **CRITICAL** The minimum required R version for this protocol is 4.4.0, and it is recommended to install the latest version.
- RStudio (<https://posit.co/download/rstudio-desktop/>)
▲ **CRITICAL** This is recommended as it is the most commonly used IDE (Integrated Development Environment) for the R language and greatly facilitates data analysis.

Example data and scripts

Download the example data (Box 3), code and a dependent package from GitHub (https://github.com/ChiLiubio/microeco_protocol_v1). To do so, click on 'Code' tab on the webpage, then download the compressed folder.

Equipment setup

Install R and RStudio. Next, create a new working directory on your computer (here we call it 'test_dir'), which will be used to store all input files and save analysis results. Then, set the working directory for R. Open RStudio, click on 'File' in the navigation bar, then select 'New Project' and choose the target folder to create a new project. If using RGui, click on 'File' in the navigation bar, then select 'Change directory' and choose the target folder. If using command

BOX 3

Input data files from example data

Amplicon sequencing data ('Input/1.Amplicon/')

- Sample_info.xlsx: an Excel file containing metadata (columns) for each sample (rows). Sample information for amplicon sequencing is located in the Excel file, including experimental treatments, sampling information, and various physicochemical properties of bulk soil. The first column, 'SampleID', is the sample name and will serve as the row names in the data frame when read into R; it should be consistent with the sample names used in bioinformatic analysis. If there is a discrepancy, a two-column data frame can be provided to the match_table parameter when using the qiime2meco function to conveniently align the sample names in the abundance table with the names in sample information table. The second column, 'Group', represents different treatment groups, namely the combinations of 'Cropping' and 'Fertilization'. 'Cropping' and 'Fertilization' are the two experimental factors. 'Cropping' denotes the experimental treatments including crop rotation and CC, and 'Fertilization' includes CK (no fertilizer control), NPK (inorganic fertilizer) and NPKS (NPK + straw amendment). For detailed experimental information, please refer to ref. 20. 'Compartment' means different sampling compartments (bulk soil, rhizosphere soil and root)
- 'QIIME2_qza': analysis result files of QIIME2 software in the 'qza' format include 'feature_table_filter.qza': ASV abundance table based on the DADA2 method; 'taxonomy.qza': taxonomic assignment table of ASV sequences based on the SILVA SSU_138 database⁷⁵; 'sequences_filter.qza': ASV sequences; and 'rooted_tree.qza': phylogenetic tree
- 'QIIME2_converted': converted files based on the 'qza' format results of QIIME2 software include 'feature_table.tsv': ASV abundance table; 'taxonomy.qza': taxonomic assignment

table; 'dna-sequences.fasta': ASV sequences; and 'tree.nwk': phylogenetic tree

Shotgun metagenomic sequencing data ('Input/2.Metagenome/')

- 'match_table.xlsx': an Excel file containing two columns, the first being the sample names in generated abundance file of bioinformatic analysis, and the second corresponding to the sample names to be replaced (that is, the sample names from the previous amplicon sequencing section)
- 'Bracken/bracken_abundance_table.txt': the analysis result file from the Bracken software⁶⁶, where rows represent taxonomic groups at various levels and columns denote individual samples
- 'HUMAN3/Metacyc_pathabundance_joint_table.tsv': MetaCyc metabolic pathway abundance data obtained from analysis with the HUMAN3 software¹⁷, with rows specifying metabolic pathways (species-specific pathways) and columns corresponding to individual samples. The abundance information in the table represents reads per kilobase (RPK)
- 'HUMAN3/KEGG_pathabundance_joint_table.tsv': KEGG metabolic pathway abundance data obtained from analysis with the HUMAN3 software, in which the format is similar to the above MetaCyc pathway data

Metabolome data ('Input/3.Metabolome/')

- 'Metabolome.xlsx': nontargeted metabolomic data is stored in an Excel sheet, with the first column listing the original IDs from the detection process, the second column providing annotation information for detected peaks and the other columns containing peak area data for samples

line mode, you can use the `setwd` function to set the R working directory. Assuming the path of the target folder 'test_dir' is 'pathfor_test_dir', the example command is: `setwd("pathfor_test_dir")`. To verify whether the directory has been set correctly, execute the command: `getwd()`. Navigate to the download directory, unzip the example data archive, and then move or copy all the folders to the aforementioned target folder ('test_dir').

Before starting the specific steps, please verify whether the R version and working directory settings are correct by following the steps below. For RStudio, click on 'File' in the navigation bar, then select 'Open File'. Locate the script 'check_working_dir.R' in the 'Rcode' folder and click the 'Source' option in the upper right corner of the window. For RGui, click on 'File' in the navigation bar, then select 'Source R code'. Locate the script 'check_working_dir.R' and confirm. For the command line mode, please run the following command: `source("Rcode/check_working_dir.R")`

Procedure

Stage 1: install R packages

● TIMING ~30 min

1. Install R packages. Use a text editor to open the 'Step1_Install_packages.R' file in the 'Rcode' folder. Alternatively, in RStudio or RGui, click on 'File' in the navigation bar, then select 'Open File' or 'Open script' to open the 'Step1_Install_packages.R' in the 'Rcode' folder. Run the commands line by line to install the required R packages. As some packages may trigger interactive prompts during installation, users should avoid running all commands at once. Instead, execute them one line at a time in an interactive R environment. If some R packages fail to install, refer to the 'Troubleshooting' section for solutions. The latter half of the code in this step is dedicated to loading the packages, which serves as a test to verify whether each package has been installed correctly. In addition to the method described above, we also provide a corresponding Docker image in which all the R packages listed in 'Step1_Install_packages.R' have been pre-installed. Users can directly pull the Docker image and utilize it without the need for additional package installations. Detailed installation and usage instructions are provided on the GitHub repository homepage (see 'Data code availability' section).

◆ TROUBLESHOOTING

Stage 2: generate microtable object for amplicon sequencing data

● TIMING ~10 min

▲ **CRITICAL** In the following steps, to analyze amplicon sequencing data, users must first generate a microtable object and perform some preprocessing. The output path for the result files is 'Output/1.Amplicon/Stage2_amplicon_microtable'.

2. Import bioinformatic analysis data and generate microtable object. The input files here are the analysis results from QIIME2, located in the folder 'Input/1.Amplicon/QIIME2_qza'. Use the `qiime2meco` function from `file2meco` package to read files with the '.qza' suffix and generate a microtable object (R6 class). Sample information, stored in an Excel file, is read into R and integrated into the microtable object in this pipeline, producing a complete dataset for subsequent analysis in R. Use the following command to do this step: `source("Rcode/Step2_Import_amplicon_files.R")`. Use the `save` function to save the microtable object in the output folder with the 'RData' suffix for data backup and easy retrieval in subsequent steps.

▲ **CRITICAL STEP** If the user's input files are of another type, an alternative, and also a more fundamental way, is to directly use the new function of the microtable class from the `microeco` package to construct a microtable object. This approach is similar to constructing the input data object using the `phyloseq` function from the `phyloseq` package. Users can refer to the help document of the microtable class to familiarize themselves with the specific format of the input data and practice the provided examples. We also

demonstrate this approach using the files in the 'Input/1.Amplicon/QIIME2_converted' folder. In such cases, the `tidy_taxonomy` function usually needs to be incorporated into the user's pipeline to unify taxonomic information. This necessity arises because different analytical software tools produce non-uniform taxonomic classifications. The function checks and supplements the prefix of taxonomic information at each classification level, as well as standardizes the naming of unclassified taxa. For example, it adds 'p_' before 'Proteobacteria' and changing 'g_unclassified' to 'g_'. Otherwise, users may encounter issues or errors for unknown reasons in some analyses associated with taxonomic abundance. To import the demonstration data, the following files generated from QIIME2 software are assumed to exist within the 'Input/1.Amplicon/QIIME2_qza' subdirectory: `feature_table_filter.qza` `taxonomy.qza` `rooted_tree.qza`.

3. Preprocess microtable object. In this step, the generated microtable object from the previous step undergoes the following processes: (1) filtering out ASVs that do not belong to the 'Bacteria' or 'Archaea' kingdoms, (2) removing ASVs annotated as 'Mitochondria' or 'Chloroplast', (3) tidying the microtable object to ensure that the sample names and ASV names are consistent across different data components inside the object, (4) renaming the ASVs in the format 'ASV + number' and (5) setting the group information in the sample information table to a factor type to facilitate orderly processing or display of groups in some subsequent analysis. Thanks to the highly encapsulated nature of the R6 class, the functions used in this step all originate from the created microtable object. Once each step is completed, the data within the microtable object are updated and no additional output is produced. We proceed with the `save` function to save the object in the output folder with the 'RData' suffix. If the sequencing was performed in batches, there may be a batch effect. We recommend using the ConQuR package⁴⁴ to correct the `otu_table` within the microtable object. Run the code as follows: `source("Rcode/Step3_Preprocess_microtable.R")`.
4. Normalization. The data normalization methods mentioned in the 'Experimental design' section can be fully implemented using the `trans_norm` class in the `microeco` package, whereas other R packages for microbiome omics data analysis do not integrate these methods as effectively. The normalized datasets are outputted separately (in 'RData' format) for users to select and use as needed. Use the following command to obtain all the normalized data: `source("Rcode/Step4_Normalization.R")`.

Stage 3: core microbiome analysis

● **TIMING** ~10 min

▲ **CRITICAL** The core microbiome refers to taxa that appear widely and are relatively abundant in samples. Studying the core microbiota is of important value for revealing stable microbial groups under certain conditions, thus providing insights into the patterns of community compositions. Although some alternative tools or customized R code can also perform the core microbiome analysis, using the `microeco` package can carry out the analysis more concisely and efficiently. The analysis results of this section are stored in the folder './Output/1.Amplicon/Stage3_coremicrobiome'.

5. Identify core ASVs for each compartment. This step identifies the core ASVs for each sampling compartment based on the rarefied data. The definition of core ASVs here is as follows: they appear in more than 50% of the samples and have an average relative abundance greater than 0.05%. The operation involves filtering samples for each compartment to produce their respective microtable objects and then using the `filter_taxa` function for computation. To do so, run the following command: `source("Rcode/Step5_Coretaxa_calc.R")`.
6. Shared and unique core ASVs among compartments. Based on the core ASVs selected for each compartment, the initial microtable object is processed to exclude the redundant ASVs, and a new microtable object is generated by merging samples from each compartment, with the sample names in this object being the compartment names. Subsequent analysis of shared and unique core ASVs is conducted using the `trans_venn` class (Fig. 2a) and visualization of the taxonomic composition in different sets is accomplished with the `trans_abund` class (Fig. 2b). To perform all these analyses, execute the following command: `source("Rcode/Step6_Coretaxa_intersection.R")`.

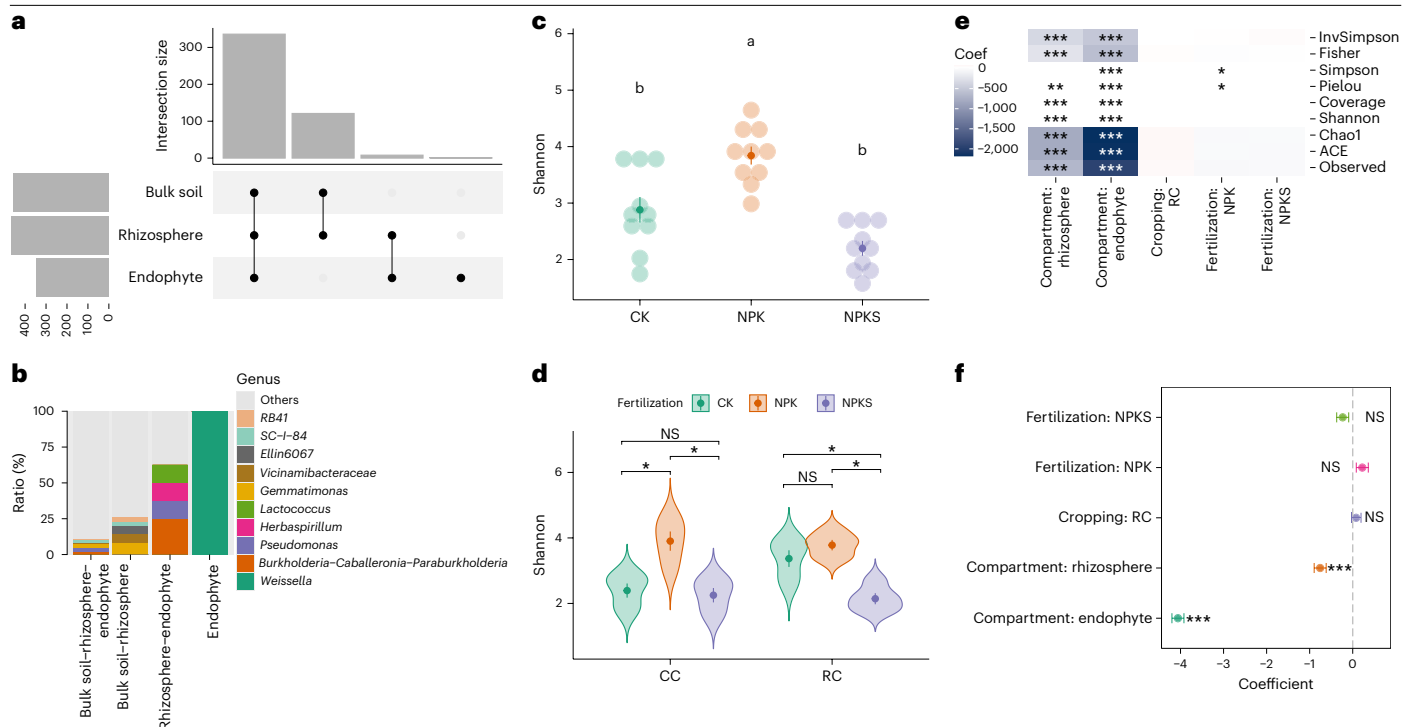


Fig. 2 | Core taxa and alpha diversity analysis. **a**, Shared and unique core ASVs in three compartments (bulk soil, rhizosphere soil and root). Top: a bar plot showing the number of each set of shared or unique core ASVs across compartments. Left: a bar plot representing the number of core ASVs for each compartment. The intersection analysis of core ASVs across different groups can reveal whether there are substantial changes in the core species under different cases. **b**, The taxonomic composition of shared or unique core ASVs across three compartments at the genus level. Each bar represents the proportion of ASVs within a group that each genus accounts for relative to the total ASVs in that group. Studying the taxonomic composition of these shared or unique core ASVs can conveniently reveal the origins of microbial groups adapted to different conditions, thereby facilitating comprehensive analysis in conjunction with functional traits and other characteristics. **c**, A dot plot of alpha diversity (Shannon-Wiener index) in root endophyte for three fertilization treatment groups. The point and error bar represent the mean and standard error, respectively. The differential test approach is the Kruskal-Wallis test with Dunn's multiple comparisons, and groups with different letters are significantly different ($P < 0.05$). **d**, The violin plot of alpha diversity (Shannon-Wiener index)

in root endophytes for three fertilization treatment groups under two cropping treatments. The differential test approach is the Wilcoxon rank sum test. The point and error bar represent the mean and s.e.m, respectively. **e**, A heatmap for coefficients (coef) and significances of linear regression fitted on multiple alpha diversity indexes. The terms 'InvSimpson' and 'Simpson' represent two expressions of Simpson's index. Assuming that D equals the sum of squared proportional abundances of species, Simpson is $1/D$; InvSimpson is $1/D$. The reference group for 'rhizosphere' and 'endophyte' is 'bulk soil'; the reference group for RC is CC; the control group for NPK and NPKS is CK. Red indicates that the index has a higher value in the treatment group compared to the control group, and blue indicates the opposite; the darker the color, the greater the change in the treatment group. **f**, The error bar of coefficients and standard errors of linear model for the Shannon-Wiener metric of alpha diversity. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, NS, $P > 0.05$. The combination of these diverse statistical methods and visualization techniques for alpha diversity demonstrates the comprehensiveness and flexibility of the relevant analytical approaches. Figure adapted from ref. 20, Springer Nature Ltd.

Stage 4: alpha diversity analysis

● **TIMING** ~5 min

▲ **CRITICAL** Alpha diversity refers to the species diversity within a community and is a crucial part of community properties. This stage calculates alpha diversity, tests the significance across groups, correlates the alpha diversity with environmental factors and visualizes the results. The output path for the result files is 'Output/1.Amplicon/Stage4_AlphaDiversity'.

- Measurement of alpha diversity. Alpha diversity can be assessed using the `cal_alphadiv` function within the microtable object. Available metrics include the observed species (ASV/OTU) number, Chao1⁴⁵, abundance-based coverage estimator (ACE)⁴⁶, Shannon-Wiener, Simpson⁴⁷, Pielou's evenness⁴⁸ and Faith's phylogenetic diversity⁴⁹. We also provide the calculation and plotting steps for rarefaction curves based on the `trans_rarefy` class in the `mecodev` package, to enable users to check whether diversity can reach a plateau based on a certain index and a given sequencing depth. We recommend using the Shannon-Wiener, Simpson, and Pielou's evenness indices, as the observed ASV/OTU number, Chao1,

- and ACE are more susceptible to the effects of sequencing depth, sequencing technique, and bioinformatic analysis methods⁵⁰. To calculate alpha diversity, run the following command: `source("Rcode/Step7_AlphaDiversity_calc.R")`.
8. Differential test and visualization of alpha diversity. Here, we use various methods implemented in the `trans_alpha` class to test differences of alpha diversity across groups and visualize the results. Univariate analysis includes the Wilcoxon test and *t*-test between two groups, and analysis of variance (ANOVA) and Dunn's Kruskal–Wallis multiple comparisons for the case of three groups (Fig. 2c). Use two-way ANOVA and regression analysis to examine the impact of different factors on alpha diversity. The case with nested groups is also covered in the analysis (such as shown in Fig. 2d). The mixed-effects model in the `trans_alpha` class can be used to analyze situations where random effects are present. Additionally, we demonstrate the Wilcoxon test and *t*-test for paired data. The examples of visualization for the statistical results of multiple factors include heat maps for multiple experimental factors and diversity indexes (Fig. 2e), and error bars of coefficients for linear models with a specified diversity index (for example, linear regression in Fig. 2f). Compared with R packages such as `phyloseq`, the statistical analysis methods covering various data scenarios and the corresponding visualization approaches are unique advantages of the `trans_alpha` class in the `microeco` package. Perform all these tests with the following command: `source("Rcode/Step8_AlphaDiversity_diff.R")`.
 9. Correlation analysis between alpha diversity and environmental factors. To associate the alpha diversity with environmental factors, use the following command: `source("Rcode/Step9_AlphaDiversity_envcor.R")`.

Stage 5: beta diversity analysis

● TIMING ~10 min

▲ **CRITICAL** Beta diversity is generally defined as the variation in species (and possibly functions or phylogeny) among communities⁵¹. Revealing the mechanisms behind these variations also falls within the scope of beta diversity research. Microbial consortia, in most cases, are driven by myriad abiotic and biotic forces that span multiple spatial and temporal scales. Studying the underlying mechanisms shaping community differences often requires the integration of environmental factor data. Methods related to the beta diversity analysis (see the 'Experimental design' section) can be divided into two major categories from a data analysis perspective: those that do not involve environmental factors and those that require environmental data. The output path for the result files in this stage is: 'Output/1.Amplicon/Stage5_BetaDiversity'.

10. Beta diversity analysis without environmental factors. First conduct beta diversity analysis without considering environmental factors. Using the distance matrix between samples to perform statistical analysis is the most common approach to studying beta diversity, as it allows for the flexible study of beta diversity by defining various distances. After calculating the Bray–Curtis and Aitchison dissimilarities, the distance matrices are subjected to PCoA (Fig. 3a,b) and non-metric multidimensional scaling. Subsequently, we present examples of PCA and DCA conducted at the genus level. In addition to ordination analysis, another frequently used approach is the transformation of distance matrix among groups, along with statistical analysis and visualization. This includes transformations of distances between samples within the same group and between different groups (Fig. 3c). The Wilcoxon test or ANOVA is then presented for differential test. Permutational multivariate ANOVA is commonly used to test differences between groups based on distance matrices⁵². Here, we show overall test among groups and pairwise tests between groups (and those under different categories) for single-factor analysis, as well as two-way analysis. As an alternative, we also conduct analysis of similarities analysis⁵³ to test the differences between groups. Carry out all these analyses with the following command: `source("Rcode/Step10_BetaDiversity_withoutEnv.R")`.

◆ TROUBLESHOOTING

11. Beta diversity analysis with environmental factors. This part of the analysis, related to environmental factors, allows users to test which factors strongly correlate to community

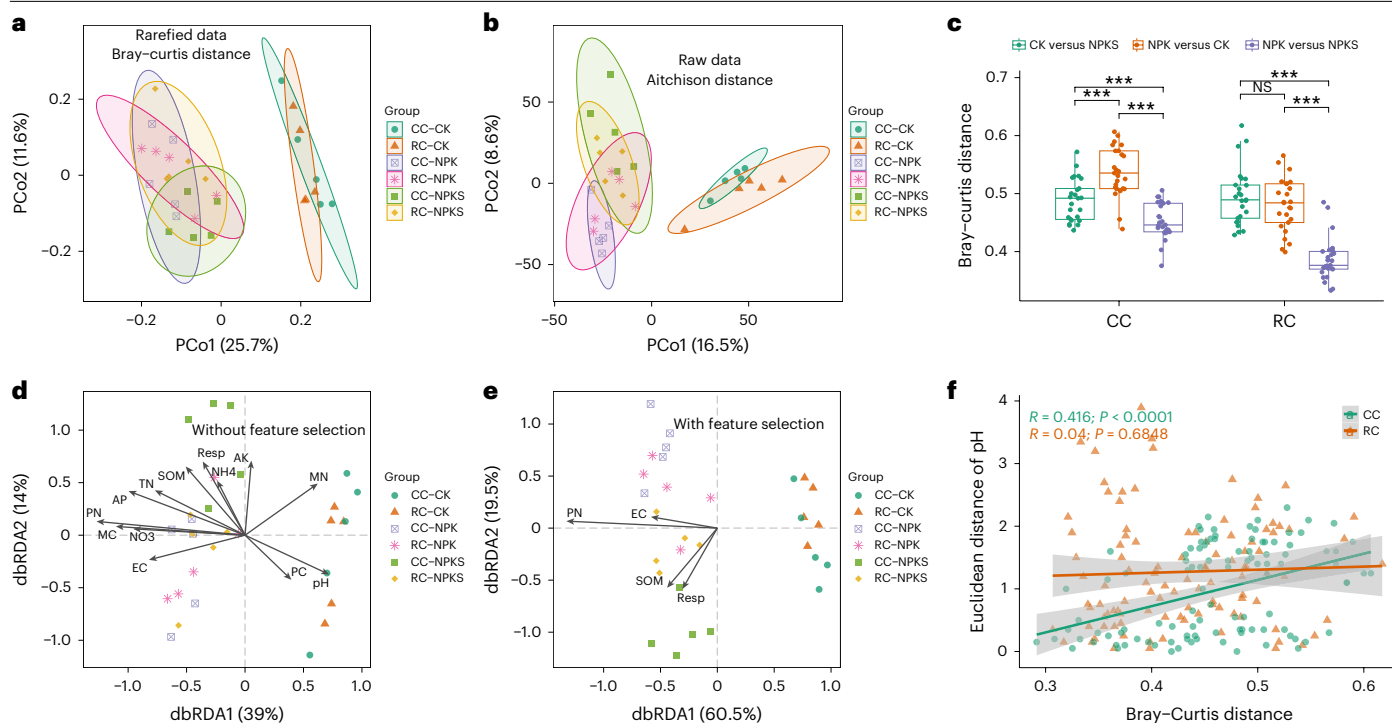


Fig. 3 | Beta diversity analysis. The beta diversity under different input matrices, calculation methods or types of analysis. **a**, PCoA based on the Bray–Curtis distance matrix, which is calculated from the rarefied dataset. **b**, PCoA based on the Aitchison distance matrix, which is calculated based on the raw dataset. **c**, A box plot of the group distance between paired fertilization treatments for CC and RC, respectively. The differential test approach is the Wilcoxon rank sum test. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, NS, $P > 0.05$. Each box represents the interquartile range (IQR), with the median depicted as a solid line within the box. The upper and lower bounds of the boxes indicate the 75th and 25th percentiles, respectively. In addition, all the values are also displayed in the form of points. **d**, The dbRDA without feature selection. The Bray–Curtis distance matrix is used

as the input data. The percentages on the axis represent the variation accounted for by the environmental variables. **e**, The dbRDA with feature selection. The optimization of the variables makes it easier to identify the main factors influencing the changes in community distances. **f**, A scatter plot of the Bray–Curtis distance and pH (Euclidean distance between samples) in rhizosphere soil for CC and RC, respectively. Pearson correlation is fitted for CC and RC separately. AK, available potassium; AP, available phosphorus; EC, soil electric conductivity; MC, microbial carbon; MN, microbial nitrogen; NH₄, ammonium nitrogen; NO₃, nitrate nitrogen; PC, plant carbon; PN, plant nitrogen; Resp, soil respiration; SOM, soil organic matter; TN, total nitrogen. Figure adapted from ref. 20, Springer Nature Ltd.

structures, thereby enabling the inference of the main drivers of community assembly. The constrained ordination methods used here include redundancy analysis (RDA), distance-based RDA (dbRDA) and correspondence analysis. Feature selection is optional for all the constrained ordination approaches (Fig. 3d,e). This advantage is not possessed by other R packages for microbiome omics data analysis. In addition, the way of selectively transforming the ordination analysis results and then visualizing them is also a unique strength of the microeco package. Additionally, the Mantel test (or partial Mantel test) analysis and scatter plot with linear correlation fitting (Fig. 3f) are shown for different groups separately. Run the code as follows: `source("Rcode/Step11_BetaDiversity_withEnv.R")`.

Stage 6: differential abundance test

● TIMING ~3.5 h

▲ **CRITICAL** Differential abundance analysis is highly complex, and many users make choices based on their limited experience in actual data analysis. For this reason, we provide analysis examples for various situations and compare some well-known methods. Compared with other R packages such as phyloseq, the methods in the `trans_diff` class of the microeco package are more comprehensive and diverse, covering a wider range of common data types. In addition, there may be multiple methods available in a certain condition, and how to choose a method for the customized data is also a challenge. Previous research has found that differential test methods

may exhibit distinct performances on a range of data types⁵⁴. It is generally left to the user to select a statistical test approach based on their limited experience. Evaluating method performance by generating simulated community profiles on specific datasets and then selecting methods more objectively holds notable value⁵⁵. Data simulation, previously common in methodological research, has rarely been adopted in other research directions. The main reasons are that data simulation is complicated⁵⁵, and using multiple statistical test methods is also challenging for users. Here, based on parametric community data simulation, we compare the power and false discovery rate (FDR) of a suite of methods and ultimately offer practical reference schemes to users. The output path for the result files is “Output/1.Amplicon/Stage6_Diff_abund”.

12. Differential abundance test at ASV level. Many approaches have been proposed for differential abundance test of ASVs/OTUs. In this step, multiple methods are demonstrated for the test at ASV level between two groups (cropping treatment) or among three groups (fertilization treatment) under a single factor, including Metastat⁵⁶, metagenomeSeq³¹, ALDEx2 (ref. 57), DESeq2 (ref. 34), edgeR³⁵, ANCOM-BC2 (ref. 58), linda⁵⁹, Maaslin2 (ref. 60), GMPR + Wilcoxon test³² and Wrench + Wilcoxon test³³. Furthermore, differential abundance tests at ASV level under dual factors are also shown. Here, the choice of method can be based on experience and existing benchmarking tests^{55,61}. When necessary, users can also refer to Steps 16–18 for data simulation and differential testing to select a more effective method. Run the following command: `source(“Rcode/Step12_Diffabund_ASV.R”)`.

◆ TROUBLESHOOTING

13. Compare the results of different methods at ASV level. The ASVs significantly detected by different approaches may vary greatly, potentially rendering the information they deliver less reliable. Therefore, it is interesting to further extract the significant ASVs obtained from different methods and explore how many ASVs are declared significant across all or a part of methods. To compare the results of various methods in the last step (for the case of three fertilization treatment groups, Fig. 4a), use the following command: `source(“Rcode/Step13_Diffabund_ASV_comp.R”)`.
14. Differential abundance test at a high taxonomic level. For the analyses at higher taxonomic levels, we use TSS normalization method to obtain relative abundance (ranging from 0 to 1). Traditional linear models are limited in their ability to model the proportional data. We thus demonstrated the examples of beta regression (betareg method)⁶² and generalized linear mixed-effects models with the beta distribution as family function (glmm_beta method; Fig. 4b). When no random effects are present, the glmm_beta method is equivalent to the betareg method. We recommend using the glmm_beta method because it is more robust and can account for random effects. Other methods used include LEfSe³⁶ and several conventional methods (ANOVA, Wilcoxon tests and linear regression) with square root arcsine transformation. Run the following code: `source(“Rcode/Step14_Diffabund_highlevel.R”)`.
15. Correlation analysis between differential taxa and environmental factors. Correlation analysis between differential taxa and environmental factors is an important method for inferring the main environmental factors driving the dynamic changes of crucial taxa. To conduct a correlation analysis, use the following command: `source(“Rcode/Step15_Diffabund_Env_correlation.R”)`.
16. Generate simulated ASV abundances for customized benchmarking. To demonstrate the benchmarking test process using customized data, we simulate community data with a given effect size in two groups using parametric method based on the SparseDOSSA2 package⁶³, which can capture the distributional and correlation structure of microbiome data. Run the following command: `source(“Rcode/Step16_Diffabund_simulate_data.R”)`.
17. Differential test for simulated data. Demonstrate the utility of differential abundance test on simulated communities by adopting the multiple approaches mentioned earlier and calculate the power (also called sensitivity) and FDR for each community with the following command: `source(“Rcode/Step17_Diffabund_simulate_test.R”)`.

◆ TROUBLESHOOTING

18. Visualization for differential test results of simulated data. Use the following command to merge the statistical analysis results of all the simulated communities and visualize the outcomes (Fig. 4c,d): `source(“Rcode/Step18_Diffabund_simulate_vis.R”)`.

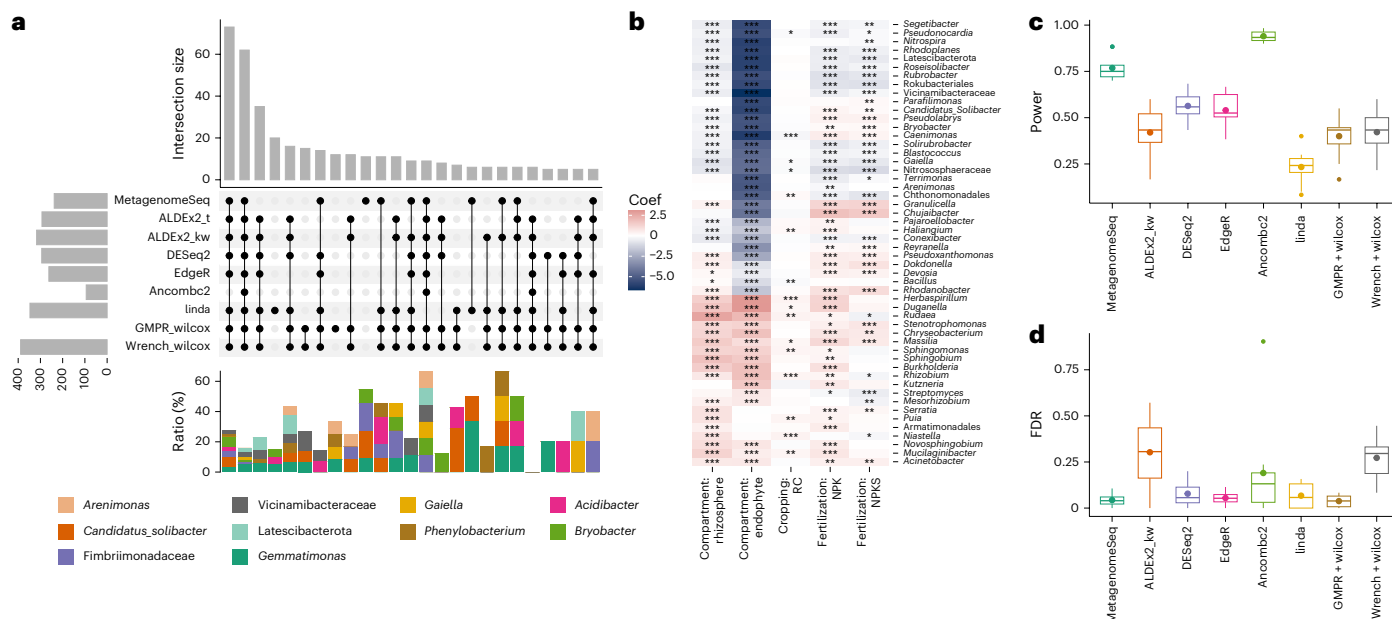


Fig. 4 | Differential abundance test. The differences in ASVs obtained by various differential abundance test methods, the patterns of relative abundance changes of genera under multifactorial treatments and the performance of each differential abundance test method under data simulation. **a**, Shared and unique ASVs significantly detected by different approaches under a single factor (fertilization treatment). Top: a bar plot showing the number of each set of shared or unique ASVs across approaches. Left: a bar plot representing the number of detected significant ASVs for each approach. Bottom: a bar plot showing the taxonomic composition in each set of shared and unique ASVs across approaches. **b**, Differential abundance test of genera based on the generalized linear model with a family function of beta distribution. Coef, the coefficient of the fitted model. *Burkholderia* represents the group of ‘*Burkholderia*–*Caballeronia*–*Paraburkholderia*’. *Rhizobium* denotes the group of ‘*Allorhizobium*–*Neorhizobium*–*Pararhizobium*–*Rhizobium*’. The reference group for ‘rhizosphere’ and ‘endophyte’ is ‘bulk soil’, the reference group for

RC is CC and the control group for NPK and NPKS is CK. Red indicates that the genus has a higher relative abundance in the treatment group compared to the control group, and blue indicates the opposite; the darker the color, the greater the change in the treatment group. * $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$. **c,d**, The power (**c**) and FDR (**d**) analysis for each differential abundance test approach based on the simulated community data. FDR is defined as $FP/(TP + FP)$, and power is defined as: $TP/(TP + FN)$, where FP is false positive, TP denotes true positive and FN is false negative. A higher power value and a lower FDR value indicate that the method performs better under the current simulated data. Each box represents the IQR, with the median depicted as a solid line within the box. The upper and lower bounds of the boxes indicate the 75th and 25th percentiles, respectively. The dot near the solid line of the median within the box represents the mean. Whiskers extend to the most extreme data points that are within 1.5 times the IQR from the box, and data points beyond this range are plotted as individual outliers. Figure adapted from ref. 20, Springer Nature Ltd.

Stage 7: machine learning models

● TIMING ~15 min

▲ **CRITICAL** In this section, we use the `trans_classifier` class to illustrate the common processes in machine learning analysis, including data splitting, data preprocessing, feature selection, model training and prediction, model evaluation, feature importance assessment and comparison of various models⁴³. We demonstrate these processes at different taxonomic levels. Compared with directly using the `caret` package and other R packages for model training and evaluation, using the `trans_classifier` class in the `microeco` package makes the entire machine learning analysis pipeline more concise and robust. The output path for the result files is ‘Output/1.Amplicon/Stage7_Machine_learning’.

19. Classification at ASV level. The CLR-normalized microtable object is selected as input data for the classification analysis of ASVs. The response variable is the ‘Cropping’ treatment in the metadata table. After splitting the data into training data and testing data, model training and prediction are performed. The random forest approach is applied as the representative of the machine learning models in this step. The default training method is tenfold cross-validation. The function `cal_feature_imp` is employed to obtain feature importance and significance. To display the model performance, receiver operator characteristic (ROC) and precision–recall (PR) curves are used for both training and testing data (Fig. 5a,b). When it is necessary to improve model performance, users can try using

- the function `cal_feature_sel` for feature selection based on the training data. This function internally encapsulates the Boruta method for feature selection⁶⁴. Perform all these processes with the following command: `source("Rcode/Step19_ML_Classifi_ASV.R")`.
20. Classification at the genus level. It is also common to perform machine learning analysis using taxa at the genus level or higher ranks. Unlike the ASV data previously mentioned, the input data at the genus level use relative abundances to illustrate the processing pipelines for two groups under cropping practice and three groups under fertilization treatment, respectively. To perform the analysis and visualize the feature importance and significance (Fig. 5c), run the following command: `source("Rcode/Step20_ML_Classifi_Genus.R")`.
 21. Compare the performance of multiple models on classification. When users are interested in multiple models, they can compare the performance of these models to facilitate selecting the best one. Use the following command to compare the classification performance of models (such as random forest, support vector machine and extreme gradient boosting; Fig. 5d) at the genus level: `source("Rcode/Step21_ML_Classifi_Genus_mulimodel.R")`.

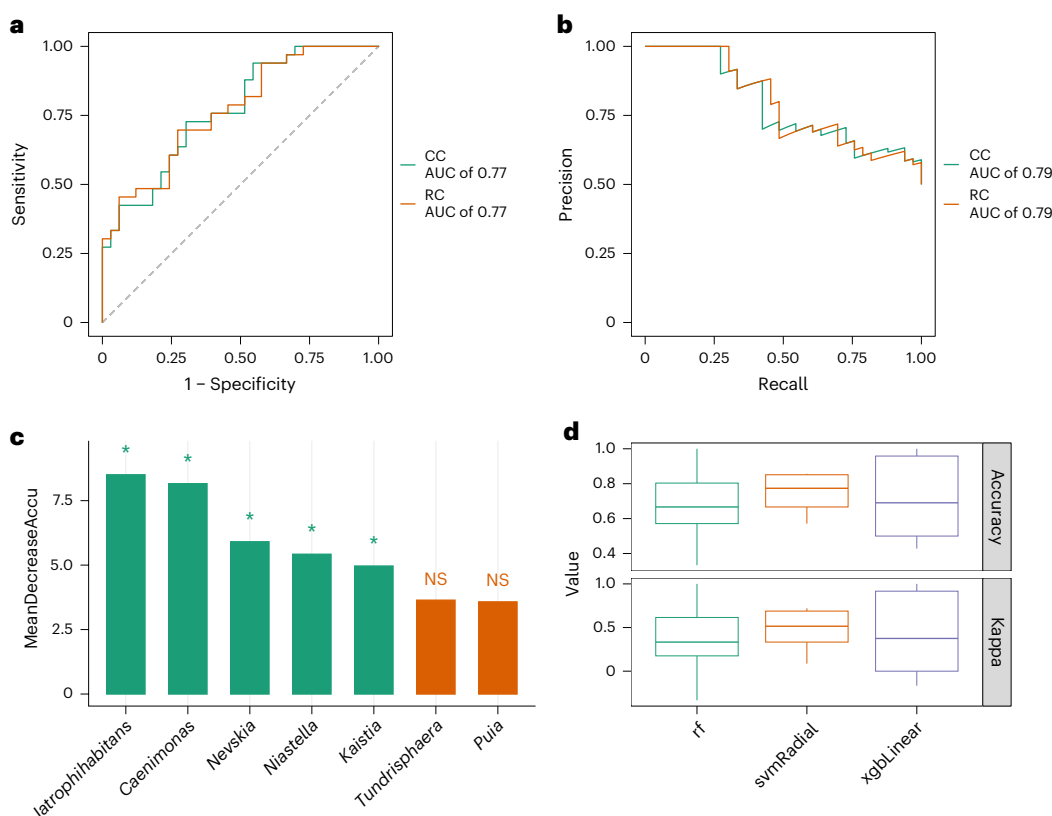


Fig. 5 | Machine learning analysis. The machine learning model performance under different evaluation metrics and analysis types, as well as the importance levels of the features. **a**, A receiver operating characteristic (ROC) curve based on the random forest classification model (cropping treatments: RC and CC) at the genus level. AUC, area under the curve. **b**, A PR curve based on the random forest classification model (cropping treatments: RC and CC) at the genus level. The PR curve assesses performance by illustrating the trade-off between precision and recall, particularly useful for imbalanced datasets. In both ROC and PR curves, higher AUC values indicate better overall model performance. **c**, The importance and significance of genera in random forest model for the classification (cropping treatments: RC and CC). MeanDecreaseAccu, mean decreasing accuracy. * $P < 0.05$; NS, $P > 0.05$. **d**, The comparison of multiple machine learning classification models based on the performance metrics (accuracy and kappa). Each box represents the IQR, with the median depicted as a solid line within the box. The upper and lower bounds of the boxes indicate the 75th and 25th percentiles, respectively. Whiskers extend to the most extreme data points that are within 1.5 times the IQR from the box, and data points beyond this range are plotted as individual outliers. rf, random forest; svmRadial, support vector machine; xgbLinear, extreme gradient boosting. Figure adapted from ref. 20, Springer Nature Ltd.

22. Regression at the genus level. When the response variable is a continuous numerical variable, the `trans_classifier` class can be conducted for the regression analysis. Here, we use pH as the response variable to exhibit the examples with the following code: `source("Rcode/Step22_ML_Regress_Genus.R")`.

Stage 8: metagenomic data analysis (HUMAN3 software)

● TIMING ~10 min

▲ **CRITICAL** As the HUMAN3 (ref. 17) software calls the MetaPhlAn software⁶⁵ to ascertain genes or metabolic pathways at the species level, the result files contain both species and gene/pathway information, making the parsing and organization of results very complex. For this reason, the `humann2meco` function has been designed in the `file2meco` package to conveniently parse the result files of HUMAN3 software. According to our experience, the `file2meco` and `microeco` packages are currently the only R analysis systems that support direct reading and analysis of data generated by the HUMAN3 software. The output path for the result files is 'Output/2.Metagenome/Stage8_HUMAN3'.

23. Import HUMAN3 analysis results. Use the `humann2meco` function from the `file2meco` package to read the MetaCyc and KEGG metabolic pathway abundance files to generate microtable objects. The function can invoke the database inside the `file2meco` package to complete the taxonomic hierarchy information of species and metabolic pathways and generate a hybrid taxonomic table, that is, `tax_table` inside the object, which can facilitate subsequent enrichment analysis. To accomplish the file import, run the following command: `source("Rcode/Step23_HUMAN3_import.R")`.
24. Statistical analysis and visualization of metabolic pathways. The results produced by the HUMAN3 software can be applied to discern which species contribute most to a group of genes or metabolic pathways, which is also the greatest difference between the HUMAN3 software and other tools. Therefore, subsequent analyses primarily focus on demonstrating this feature, while also showing some other conventional analysis contents, including the visualization of overall abundances of metabolic pathways (Fig. 6a),

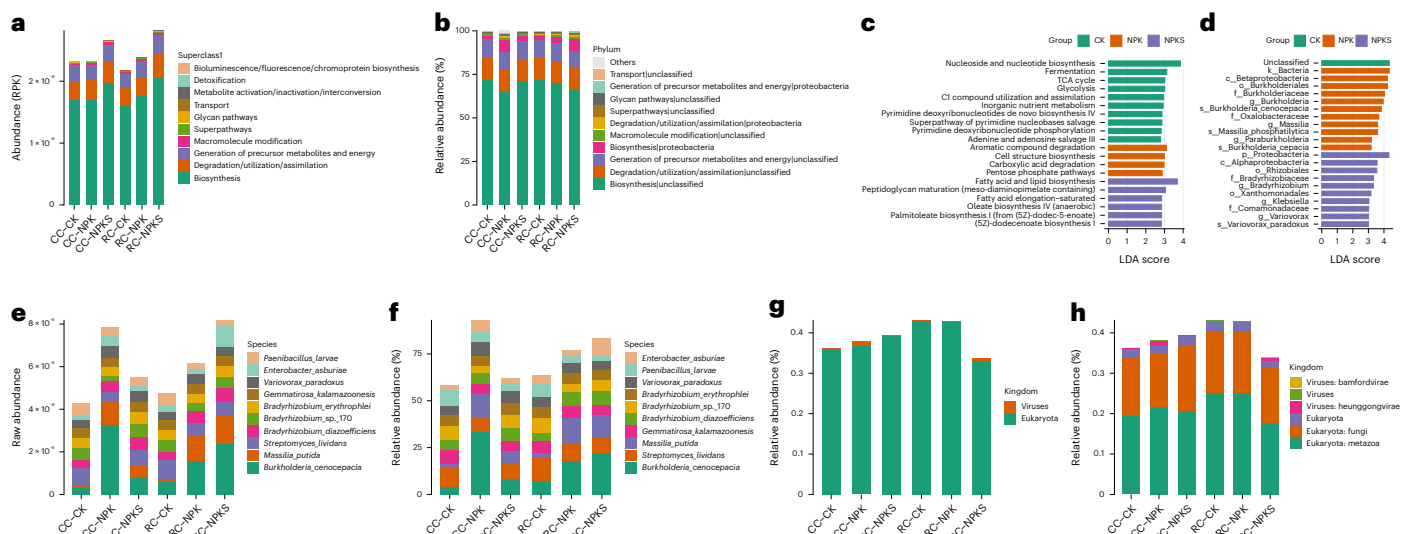


Fig. 6 | Metagenomic data analysis. The abundance and importance of metabolic pathways or taxa under different parameter choices. **a**, The mean relative abundance of all MetaCyc pathways in dominant genera across treatment groups. **b**, The mean relative abundance of MetaCyc pathways in different phyla across treatment groups. **c**, The linear discriminant analysis (LDA) score in LEfSe for MetaCyc pathways across fertilization treatments. The higher the LDA score, the greater the effect this metabolic pathway has on distinguishing between different groups in the model. **d**, The LDA score in LEfSe for taxa contributing to

the detected MetaCyc pathways across fertilization treatments. **e**, The mean raw abundance of dominant species across treatment groups detected by Bracken software. **f**, The mean relative abundance of dominant species across treatment groups based on Bracken analysis. **g**, The mean relative abundance of eukaryotes and viruses across treatment groups based on Bracken analysis. **h**, The mean relative abundance of eukaryotes and viruses with duplicated prefixes at the kingdom level across treatment groups based on Bracken analysis. Figure adapted from ref. 20, Springer Nature Ltd.

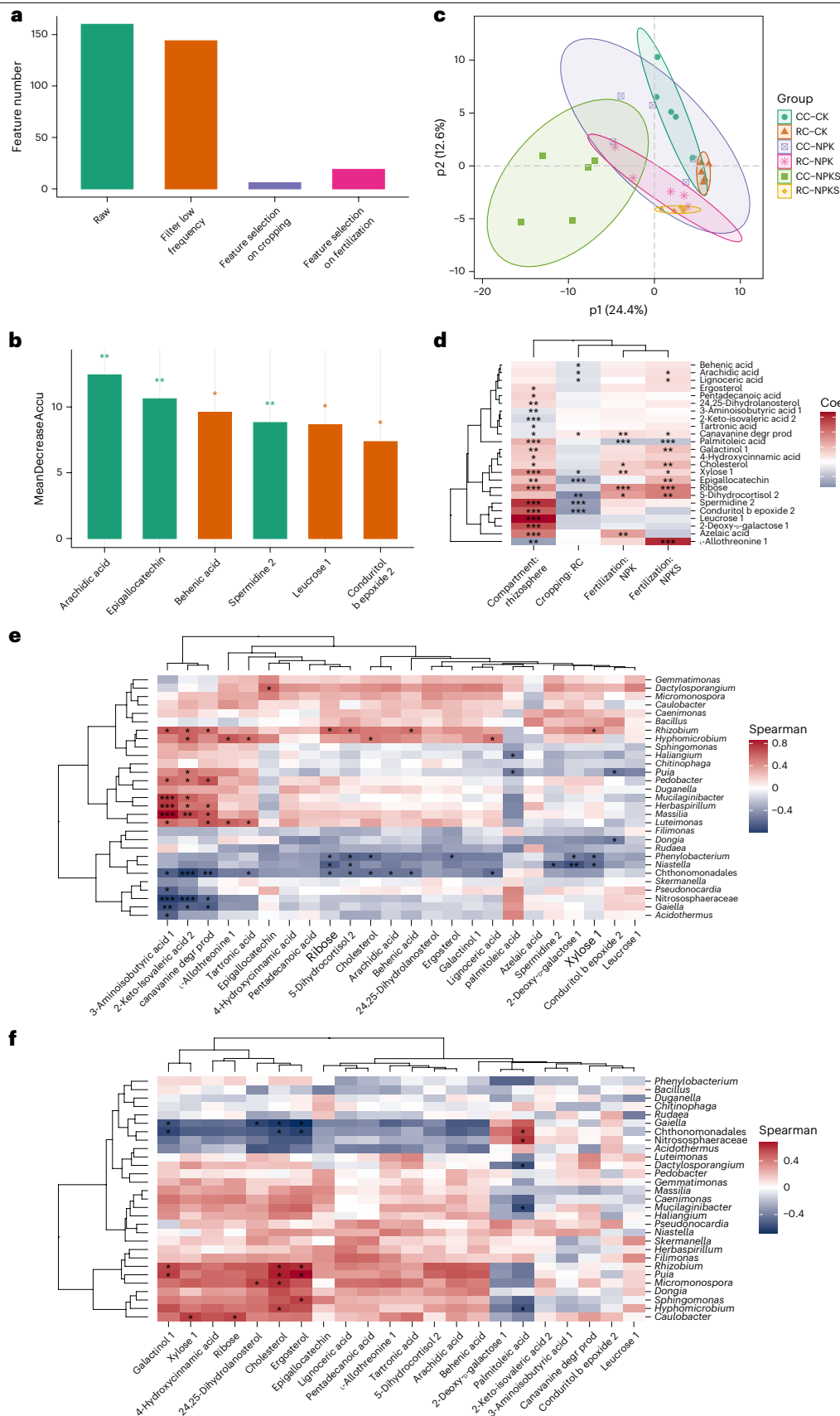


Fig. 7 | Metabolomics data analysis. The results of different analytical methods applied to metabolomics data, aiming to identify significant metabolites and explore their variation patterns across treatments as well as their relationships with microbial taxa. **a**, Feature numbers in the data after each analysis step, including the input data, filtering features with low occurrence frequencies and feature selection for cropping and fertilization treatments. **b**, The importance and significance of biomarkers (selected metabolites) in a random forest model for the classification (cropping treatments). MeanDecreaseAccu, mean decreasing accuracy. **c**, The visualization of scores in partial least squares discriminant analysis for metabolites in rhizosphere soil. **d**, Linear regression with log-transformed metabolic data. The features are selected based on the

random forest models for cropping and fertilization treatments, respectively. The reference group for 'rhizosphere' is 'bulk soil', the reference group for RC is CC and the control group for NPK and NPKS is CK. Red indicates that the metabolite has a higher abundance in the treatment group compared to the control group, and blue indicates the opposite; the darker the color, the greater the change in the treatment group. **e**, Spearman correlation analysis between selected metabolites and genera in rhizosphere soil samples. **f**, Spearman correlation analysis between selected metabolites and genera in bulk soil samples. The *P* values were adjusted using the Benjamini–Hochberg FDR approach. **P* < 0.05, ***P* < 0.01 and ****P* < 0.001. degr prod, degradation product. Figure adapted from ref. 20, Springer Nature Ltd.

visualization of metabolic pathway abundance at a specific microbial taxonomic level (Fig. 6b), differential abundance test of metabolic pathways (Fig. 6c) and differential abundance test of taxa contributing to the identified metabolic pathways (Fig. 6d). In these analyses, the application of the `cal_abund` function within the `microtable` object for obtaining enriched abundance is key to data transformation. Owing to its design for orderly enrichment calculations according to selected levels, it is suitable for processing data with complex taxonomic and functional information. Users can choose to focus solely on metabolic pathways (or genes), solely on taxa or on a combination of both. When selecting a combination of metabolic pathways and taxa, the user can assess the contributions of microbial guilds to the abundance of each pathway. In addition, functional alpha diversity based on pathway abundance is also evaluated. In these examples, either the original normalized abundance (reads per kilobase) or the relative abundance generated by the TSS method was used for analysis. We do not recommend using other methods (such as CLR or CSS) to normalize the data as the compositional effect is relatively weak in these data compared to amplicon sequencing data. Conduct all these analyses with the following command: `source("Rcode/Step24_HUMAN_analysis.R")`.

Stage 9: metagenomic data analysis (Kraken2/Bracken software)

● TIMING ~10 min

▲ **CRITICAL** The primary difference between the results format of Kraken2 and amplicon sequencing analysis software (for example, QIIME2) is that amplicon sequencing analysis aggregates the abundance of taxa from the most refined ASV/OTU level to higher levels, so the abundance of a taxon equals the sum of the abundances of all taxa at more refined level it contains. By contrast, the results of Kraken2 results do not have this direct correspondence, and all abundances at each taxonomic level are merged into one file. Furthermore, the Bracken software⁶⁶ can be used to estimate the abundance of species or other taxa based on the results of Kraken2. It shares the same result format as Kraken2, with differences in some abundance values. The output path for the result files is 'Output/2. Metagenome/Stage9_Bracken'.

25. Import Bracken software results. The `mpa2meco` function in the `file2meco` package is specifically designed to read abundance files generated from Kraken2 and Bracken software. The `otu_table` in the resulting `microtable` object defaults to the data from species level. The original abundances at various taxonomic levels are organized in the `taxa_abund` list, which contains multiple data frame objects. Therefore, rerunning the `cal_abund` function in the `microtable` object will overwrite the data in this list. The parameter `rel = TRUE` can be used to set the data in the list to relative abundance (where the sum of all features equals 1). As some results may have multiple classification information at the kingdom level, such as 'k_Eukaryota|k_Fungi|p_Ascomycota' and 'k_Viruses|k_Heunggongvirae|p_Uroviricota', the function includes a `sel_same` parameter to choose whether to keep the first, second or both when generating the `tax_table`. When choosing to keep both, the '|' is converted to ':' to avoid affecting other subsequent analyses. For the kingdom-level abundance in the raw `taxa_abund` list, both are retained. To import

example data from Bracken analysis, use the following command: `source("Rcode/Step25_Bracken_import.R")`.

26. Statistical analysis and visualization on imported abundance data. Due to the generated microtable object being very similar to the format of amplicon sequencing data, the subsequent analysis processes for Kraken2/Bracken data are also quite similar to that in previous amplicon sequencing data sections. Here, some relatively routine analyses are performed. We also carried out some procedures to underscore the data characteristics of Bracken results, such as the visualization of raw and relative abundances (Fig. 6e,f), and the visualization of kingdom abundances with different prefix options (Fig. 6g,h). Carry out the following command: `source("Rcode/Step26_Bracken_analysis.R")`.

Stage 10: metabolomic data analysis

● **TIMING** ~20 min

▲ **CRITICAL** Microorganisms interact with hosts (for example, plants and animals) primarily through metabolites as mediators^{67–69}. Both the host and microorganisms can encode diverse metabolic pathways to produce a wide spectrum of metabolites, which makes the study of these interactions extremely complex. Metabolomics methods are of great value in elucidating the metabolic mechanisms. In this section, we have selected different types of methods from the microeco package for the analysis of the example soil metabolomic data.

27. Import metabolomic data. The metabolomics abundance data is typically in the format of a sample-feature table, allowing for direct data reading using general R functions. Here, the untargeted metabolomic data stored in Excel is directly imported to generate a microtable object, which is then preprocessed and saved locally in 'RData' format. Use the following command: `source("Rcode/Step27_Metabolome_input.R")`.
28. Machine learning for metabolomic data. This step involves establishing and predicting a machine learning model based on metabolite data, and obtaining the importance and significance of each selected metabolite. The response variables are selected from the sample information table for 'Cropping' and 'Fertilization' columns separately. Metabolites that appear with an occurrence frequency lower than 20% in the samples are filtered out before model building. We use the function `cal_preProcess` to standardize and centralize feature abundance and to remove features with extremely low variance in the training data⁷⁰. The number of remaining metabolites in each step is counted and presented (Fig. 7a). The names of important metabolites determined in the feature selection step are saved locally in 'RData' format for convenience in subsequent analyses (for example, these features in Fig. 7b). In addition to using the random forest method in the `trans_classifier` class, we have also demonstrated the partial least squares discriminant analysis method implemented in the `trans_beta` class (Fig. 7c). To perform these operations, run the following code: `source("Rcode/Step28_Metabolome_classifier.R")`.
29. Differential test of metabolomic data. This step focuses on the differential test and visualization of the important metabolites identified in the previous step, examining the abundance changes and significance of metabolites in the treatment groups (Fig. 7d). A log transformation is applied to the metabolite abundances to reduce intersample variability. To do so, use the following command: `source("Rcode/Step29_Metabolome_diff.R")`.
30. Correlation analysis between metabolites and genera. Using some of the significant metabolites identified earlier, a correlation analysis is conducted with the relative abundances of microbial taxa at the genus level to unravel the association between microbes and metabolites. Although the inference from correlation analysis may be obscured by metabolite properties, this method is still valuable for uncovering the relationships between metabolites, microorganisms and their habitats. The selection of prokaryotic genera is performed based on the results of beta regression analysis. Spearman correlation analysis and network analysis of metabolites and genera are conducted separately for rhizosphere and non-rhizosphere soil (Fig. 7e,f). To finish this step, run the following command: `source("Rcode/Step30_Metabolome_taxa_cor.R")`.

Troubleshooting

Troubleshooting advice can be found in Table 1.

Table 1 | Troubleshooting table

Step	Problem	Possible reason	Solution
1	Failed to download a package. Error in download.file(url, destfile, method, mode = "wb", ...); Timeout of 60 s was reached	The internet speed is too slow	Two approaches: the first is to increase the allowable download time; example code is as follows: options(timeout = 1000); the second is to attempt to change the resource download mirror. First, find the nearest mirror to the user at https://cran.r-project.org/mirrors.html , and then modify the CRAN and Bioconductor mirrors in R with the following example code: mirror = " https://mirrors.tuna.tsinghua.edu.cn "; options("repos" = c(CRAN = paste0(mirror, "/CRAN/")));options(Bioc_mirror = paste0(mirror, "/bioconductor/"))
1	Failed to install Bioconductor packages. Error: Bioconductor version cannot be validated; no internet connection?	Secure socket layer (SSL) setting issue	(1) Add CURL_SSL_BACKEND=openssl to the .Renviron file (https://csgillespie.github.io/efficientR/set-up.html) or computer environment variable (variable: CURL_SSL_BACKEND; value: openssl); (2) add options(download.file.method="libcurl", url.method="libcurl") to the .Rprofile (https://csgillespie.github.io/efficientR/set-up.html) or Rprofile.site file (under R_installation_path/etc/); (3) restart R
10	Failed to calculate UniFrac distance. Error in test\$cal_betadiv(unifrac = TRUE): No phylogenetic tree provided	The phylogenetic tree is not available in some customized analysis	Set "unifrac = FALSE" in cal_betadiv function
12, 17	Failed to run ancombc2 method. Error in model.matrix.formula(...): data must be a data.frame	The originally called function model.matrix from the stats package has been replaced by a similar function from another loaded package	Install the ANCOMBC package with a version ≥2.8.1, or restart R and rerun this script
14	Failed to plot cladogram of LEfSe. There is no result figure output, or some unknown errors occur	The format of taxonomic information is chaotic	Try to run the following code: your_microtable_object\$tax_table %<>% tidy_taxonomy; your_microtable_object\$cal_abund()

Timing

The software installation time is primarily determined by internet speed, and the duration required for data analysis mainly depends on computer performance (for example, CPU performance), and is also largely affected by the user's data analysis habits. Simply running through all the processes will take at least 5 h. Some steps are particularly time consuming, such as Steps 16 and 17, which involve community data simulation and differential abundance testing. Although it can be time consuming, we strongly recommend that users take the time to understand the purpose of each step in the code and review the help documents of functions in the corresponding class.

Anticipated results

If practitioners follow each step of this protocol correctly, all analysis results and figures, including those displayed in this article (Figs. 2–7), will be saved in the respective output folder for each stage. Detailed explanations of the output results are provided in the annotation lines of the code for each step. Additionally, when the output includes both files and graphs, these will be correspondingly related, as demonstrated in the operational steps. Users may choose to skip certain steps if desired; however, if the program requires input data from steps that have not been executed, it will generate an error message. Users can then run the necessary steps according to the instructions. The code in this protocol is designed to be easily adaptable to

a wide range of experimental designs. We also encourage users to make their modified code publicly available when publishing research results to enhance reproducibility and provide convenience for others.

Data availability

All the input data used in the protocol are freely available on GitHub (https://github.com/ChiLiubio/microeco_protocol_v1) and Gitee (https://gitee.com/chiliubio/microeco_protocol_v1) repositories.

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Author contributions

C.L., X.L. and M.Y. conceived the study and wrote the paper. C.L. and F.R.P.M. wrote all the code. C.L. created all the figures. H.L., A.B.V. and R.J.Z. tested all the steps in the protocol and improved the draft writing.

Competing interests

The authors declare no competing interests.

Additional information

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